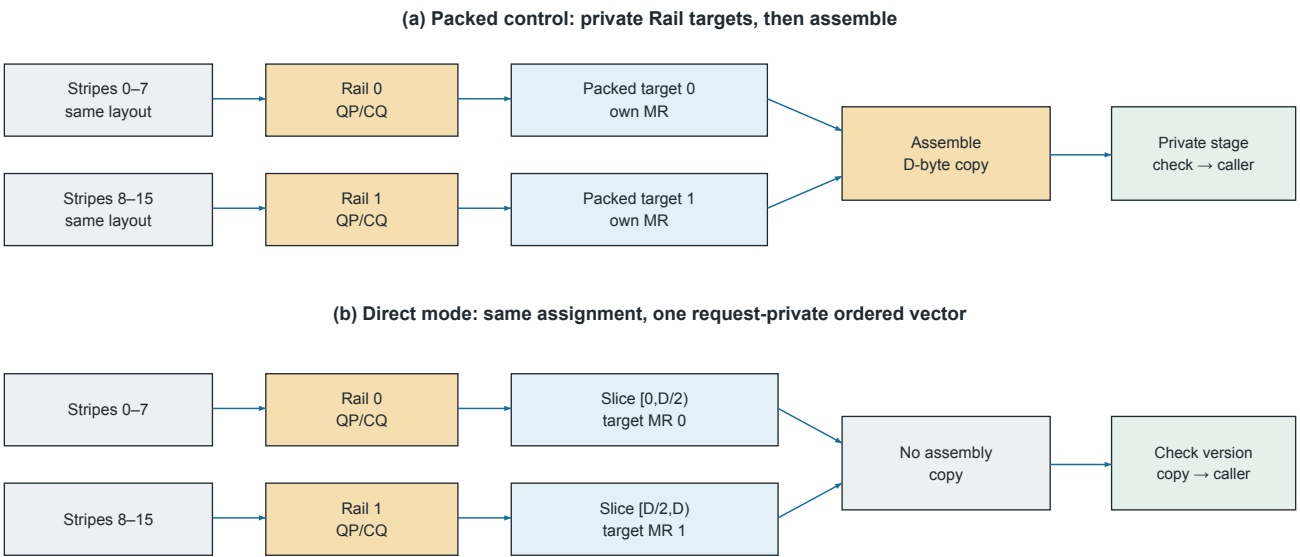

(a) Packed control: private Rail targets, then assemble
Stripes 0–7
same layout
Rail 0
QP/CQ
Packed target 0
own MR
Stripes 8–15
same layout
Rail 1
QP/CQ
Packed target 1
own MR
Assemble
D-byte copy
Private stage
check → caller
(b) Direct mode: same assignment, one request-private ordered vector
Stripes 0–7
Rail 0
QP/CQ
Slice [0,D/2)
target MR 0
Stripes 8–15
Rail 1
QP/CQ
Slice [D/2,D)
target MR 1
No assembly
copy
Check version
copy → caller
Both: QP retired before MR release; private stage held through checks and publish/fail